

Language Technologies for Crisis Preparedness and Response (LT4CPR)

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1 Workshop Topic and Content

The world is entering an era of increasing frequency and severity of crises, including an international pandemic and climate-related natural disasters. In addition, there are many ongoing armed conflicts, leading to mass displacements and regional crises. A range of mitigating actions includes disaster preparedness, disaster risk reduction, and when all else fails, crisis response.

Speech and language technology (SLT) offers a broad range of options in crisis scenarios, such as text classification for triaging tweets for immediate assistance following the Haiti earthquake in 2010; machine translation for COVID-19 information from the World Health Organization; automated vote monitoring for a referendum in Southern Sudan; a UN-led earthquake simulation in Colombia, which included translation and triaging. Often, the regions most affected by crises also suffer from entrenched disadvantage and political upheaval, which means that we need to rely on informal communication channels and locally prevalent languages.

The matter of linguistic diversity can be especially acute, as communication may happen in regional languages which lack a standardised orthography, or local spoken varieties of international languages, none of which are supported in technology. How do we leverage SLT in the presence of this linguistic diversity, especially in scenarios that generally require rapid response?

Tagline: *Develop language technologies to prepare for and respond to crises—bridging NLP, local languages, and real-world humanitarian impact.*

We propose to organize a 1-day workshop on Language Technologies for Crisis Preparedness and Response (LT4CPR). We anticipate that this workshop will be the first of a series, and be the natural home for NLP work on emerging crises, such as what motivated the once-off *Workshops on NLP for COVID-19* (Verspoor et al., 2020a,b).

We will also host a panel discussion on language technologies in crisis preparedness and response, focusing on strategic priorities and on possible

shared infrastructure. We have laid the groundwork for this panel by conversing with people in allied fields (see below for invited speakers).

The success of the NLP4Covid workshops and the recent revival of the NLP for Positive Impact workshop shows the desire of the NLP community to build language technologies for crisis/disaster response that will be useful for everyone. However, this needs to be done in coordination with all other stakeholders who work in the crisis response area.

2 Invited Speakers

The workshop will seek to shape the vision of the SLT community for the future, informed by experts in crisis preparedness and response, and include people who have already been bridging these areas. Thus, we aim to hold one keynote speaker and a panel discussion. **Travel for the invited speakers will be provided through the proposer’s NSF CIRC award that supports this workshop.**

We have reached out to a diverse pool of notable crisis informatics and related scholars, most of who have enthusiastically tentatively accepted to participate:

- **Jonatan Lassa** has already agreed to be one of the speakers. He has a PhD from the United Nations University, followed by postdoctoral fellowships at Harvard’s Ash Center and at the Institute of Catastrophic and Risk Management at Nanyang Technological University. He currently holds a position in disaster governance and emergency management at Charles Darwin University. Jonatan has ongoing research interests in disaster governance, post-disaster need assessment, climate adaptation, and food security.
- **Manuela Noske** who works for ClearGlobal (formerly Translators without Borders), providing on-the-ground LT in humanitarian response;
- **Leysia Palen** (Univ. of Colorado at Boulder), one of the most cited scholars in HCI for crisis scenarios;
- **Sharon O’Brien** (Dublin City Univ, Computer Science), who has worked on translation

in crisis scenarios;

- **Kalika Bali** (Microsoft Research India) who works on LT for local languages and the global south.

3 Workshop Size / Prior Events

- This is the first iteration of workshop on this topic at an NLP conference.
- No special requirements and technical needs beyond a standard hybrid setup for remote participation and video recordings.
- Workshop duration: 1 day.

4 Shared Task

We will include a shared task on a *novel task*: **automatic situation report generation**, based on social media content pertaining to specific crisis scenarios. A crucial task during a crisis is to summarize information on the ground for consumption by aid providers. Aid agencies commonly rely on situation reports, “sitreps”, usually compiled by hand, and issued initially within the first 24 hours of a crisis, and then daily afterwards (this follows UN guidelines). Automating the process, even partially, could bring immense real-world value.

We estimate 5-10 participant teams in this Shared Task.

Preferred Venue: Both EMNLP and ACL are acceptable venues.

5 Diversity and Inclusion

Contribution to academic diversity: Our goals in this workshop are larger than the workshop itself, since we aim to bring together communities and researchers with diverse backgrounds to address the problems of applying SLT to crisis scenarios. In particular, the workshop will aim to bring communities together so as to develop a shared vision and the partnerships required for an innovative research direction that will bring together computer scientists and computational linguists (focusing on SLT, especially machine translation, speech recognition, and spoken language translation) and experts in crisis preparedness and response, human-computer interaction, as well as linguists working with and representatives from local speech communities, aid agencies, NGOs, and relevant governmental bodies. The goal is to motivate and facilitate further interactions between these diverse communities, and

catalyze a robust research community focused on language technologies for crisis preparedness and response.

We will particularly encourage papers that describe situations in the crisis space for non-dominant languages, and will include invited presentations, e.g., for researchers outside of NLP and LT, where the value proposition is an invited talk instead of a publication.

Diversifying representation: The workshop will particularly encourage submissions from underserved communities that are admittedly the ones most under threat by crises.

Our selection of invited speakers and panelists includes academics from 3 continents, as well as members of relevant industries, including two that work in the Global South. Most of them have not appeared as panelists or keynote speakers in NLP conferences.

6 Workshop Organizers

Antonios Anastasopoulos antonis@gmu.edu
George Mason University, USA

Antonios Anastasopoulos is an assistant professor in the Computer Science department at George Mason University. He obtained his Ph.D. in Computer Science from the University of Notre Dame, focusing on natural language processing for endangered languages. His projects and research revolve around NLP for endangered language documentation and preservation, cross-lingual transfer learning and multilingual models for machine translation and speech recognition, as well as model robustness. In 2020-2021 he also worked on the Translation Initiative for COVID-19 (TICO-19) ([Anastasopoulos et al., 2020](#)), a cross-industry and academic initiative for developing resources to support aid response in the COVID-19 pandemic. He has served *ACL and other workshops at various posts and co-organized more than 5 *ACL workshops in the past years.

William Lewis wlewis2@uw.edu
University of Washington, USA

William Lewis is an affiliate assistant professor in the Linguistics Department at the University of Washington. He obtained his Ph.D. in Linguistics, with an emphasis in Computational Linguistics, from the University of Arizona at Tucson. His primary research interests are in endangered and threatened language preservation through the

development of resources and tools, the Online Database of Interlinear text (ODIN) (Lewis and Xia, 2010) likely the most well known of these. His interest in this space, in collaboration with the rest of the Microsoft Translator team, led to the development of the world's first production model for Haitian Kreyòl, in response to the earthquake in Haiti (Lewis, 2010). He was one of the principal researchers in defining the Crisis MT Cookbook (Lewis et al., 2011), a model for repeating the success of the Haitian Kreyòl initiative, but to be able to do so for any of the world's languages. More recently, he has worked on the Translation Initiative for COVID-19 (TICO-19) (Anastasopoulos et al., 2020), a cross-industry and academic initiative for developing resources to support aid response in the COVID-19 pandemic. He has served *ACL and other workshops at various posts and co-organized more than 10 *ACL workshops in the past 15 years.

Fei Xia fxia@uw.edu

University of Washington, USA

Fei Xia is a full professor in the Linguistics Department at the University of Washington. She obtained her Ph.D. in Computer Science from the University of Pennsylvania. Her research covers a wide range of NLP topics including basic NLP pipeline (e.g., word segmentation, POS tagging, parsing, named-entity recognition), treebank development, machine translation, and information extraction. One of her research interests is to develop resources and NLP systems for low-resource languages. For instance, she collaborated with William Lewis in building the Online Database of Interlinear text (ODIN) (Lewis and Xia, 2010) which consists of interlinear glossed text for thousands of languages automatically extracted from linguistic documents. She collaborated with Emily Bender and others in automatically building grammars, answering linguistic questions, and bootstrapping parsers using the ODIN data. She has served *ACL and other workshops at various posts and co-organized more than 10 *ACL workshops in the past 20 years.

7 Program Committee

The program committee of the LT4CPR workshop will feature researchers with expertise in a diverse set of topics related to our theme, including crisis

informatics, crisis communication and translation, technology in disaster management, disaster risk reduction and management, and multilingual NLP for local/low-resource languages. Our proposed list of program committee members is given below; not all listed have been contacted.

Kenneth Mark Anderson, CU Boulder, USA
Kalika Bali, Microsoft, India
Laurent Besacier, Naver Labs, France
Jake Rom Cadag, UP Diliman, Philippines
Zi-Yi Dou, CMU, USA
Akiko Eriguchi, Microsoft, USA
Federico Federici, UCL, UK
Marcello Federico, Amazon, USA
Christian Federmann, Microsoft, Germany
Orhan Firat, Google, USA
Stephanie Foster, MIT Lincoln Lab, USA
Francisco Guzmán, Meta, USA
Hany Hassan, Microsoft, USA
Annelies Heijmans, VHL, Netherlands
Dorothea Hilhorst, EUR, Netherlands
Junjie Hu, CMU, USA
MacDuff Hughes, Google, USA
Muhammad Imran, QCRI, Qatar
Philipp Koehn, JHU, USA
Jonatan Lassa, CDU, Australia
Patrick Meier, WeRobotics, USA
Robert Munro, Apple, USA
Dewald van Niekerk, NWU, South Africa
Manuela Noske, Translators without Borders, USA
Sharon O'Brien, DCU, Ireland
Kiyonori Ohtake, NICT, Japan
Alp Öktem, Translators without Borders, Spain
Leysia Palen, CU Boulder, USA
Alexis Palmer, CU Boulder, USA
Emmanuel Raju, KU, Denmark
Christian Reuter, TU Darmstadt, Germany
Steve Richardson, BYU, USA
Jeannette Sutton, SUNY Albany, USA
Irina Temnikova, GATE, Bulgaria

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